# Distal Arthrogryposis Type 2B1 (DA2B1 / Sheldon-Hall Syndrome): A Comprehensive Disease Report

**Target disease:** Distal Arthrogryposis Type 2B1 (DA2B1) **MONDO ID:** MONDO:0020820 | **OMIM:** 601680 | **Causal gene: TNNI2** (fast-twitch skeletal-muscle troponin I; HGNC:11946; NCBI Gene 7136; UniProt P48788; gene OMIM 191043; 11p15.5)

This report concerns the **TNNI2**-caused form of Sheldon-Hall syndrome. The "2B" distal-arthrogryposis phenotype is genetically heterogeneous and can also arise from *TNNT3* (DA2B2), *MYH3* (DA2B3), and *TPM2* (DA2B4); the entry addressed here is the **TNNI2** allelic form (DA2B1), following the MONDO ID and causal gene provided.

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## Summary

**Distal arthrogryposis type 2B1 (DA2B1), also known as Sheldon-Hall syndrome (SHS), is a rare autosomal-dominant congenital contracture disorder caused by heterozygous gain-of-function mutations in *TNNI2***, the gene encoding the fast-twitch skeletal-muscle isoform of troponin I at chromosome 11p15.5. Although the broad "DA2B" phenotype is genetically heterogeneous — mutations in four interacting sarcomeric/fast-twitch genes (*TNNI2*, *TNNT3*, *TPM2*, *MYH3*) can all produce a Sheldon-Hall picture — the *TNNI2*-driven form is DA2B1 specifically. The clinical hallmark is non-progressive distal limb contractures (camptodactyly, ulnar deviation of the hands, clubfoot/congenital vertical talus) combined with a distinctive triangular face (downslanting palpebral fissures, prominent nasolabial folds, small mouth, prominent chin). Cognition and life expectancy are normal.

The molecular mechanism is **gain of function, not haploinsufficiency**. DA2B1 *TNNI2* mutations cluster in the troponin-I C-terminal (inhibitory/actin-binding) domain and increase the calcium sensitivity and contractile output of developing fast-twitch fetal muscle. In reconstituted actomyosin and skinned-fiber assays, the mutant proteins raise ATPase rates and $Ca^{2+}$ sensitivity, and the *TNNI2* R156ter mutant generates higher maximum force. Sustained hypercontractility of fetal muscle during the critical window of limb development is believed

to restrict joint movement in utero, producing fixed contractures. A second, mouse-demonstrated branch links a *Tnni2* mutation to a non-sarcomeric transcriptional route in bone (increased *Hif3a*, reduced *Vegf*, delayed endochondral ossification) that contributes to short stature.

This report synthesizes **7 confirmed findings** and **28 reviewed papers** into a complete disease-knowledge-base entry covering identifiers, etiology, phenotypes, genetics, mechanism, anatomy, temporal course, epidemiology, diagnostics, prognosis, treatment, prevention, cross-species biology, and model organisms. Evidence derives from human clinical/genetic studies, in-vitro biophysics, and a knock-in mouse model. Treatment is entirely supportive (physical/occupational therapy, Ponseti casting, staged orthopedic surgery); there is no disease-specific or curative drug therapy.

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## Key Findings

### Finding 1 — DA2B1 is caused by heterozygous gain-of-function *TNNI2* mutations clustered in the C-terminal domain at 11p15.5

*TNNI2* encodes the fast-skeletal troponin I isoform, one of four sarcomeric/fast-twitch genes that cause distal arthrogryposis. In the Beck et al. (2013) cohort of 153 distal-arthrogryposis kindreds, disease-causing mutations in 56/153 (37%) families — including 14/48 (29%) with DA1 and 42/105 (40%) with DA2B — were distributed **nearly equally across *TNNI2, TNNT3, TPM2, and MYH3***. Reported *TNNI2* variants are heterozygous and concentrate in the **C-terminal domain**: p.Lys175del (c.523_525delAAG), p.Arg156Ter (c.466C>T), p.Glu167del, p.Arg174Gln, and p.Ile165Phe (c.493A>T). Inheritance is autosomal dominant with variable expressivity and reduced penetrance.

> "Disease-causing mutations in 56/153 (37%) kindreds including 14/48 (29%) with DA1 and 42/105 (40%) with DA2B were distributed nearly equally across TNNI2, TNNT3, TPM2, and MYH3" — PMID: 23401156
>
> "detected a three base pair deletion that would be predicted to result in the deletion of a glutamic acid at codon position 167 (DeltaE167). This mutation, like the two previously described TNNI2 mutations, is located in the carboxy-terminal domain" — PMID: 16497570

## Finding 2 — DA-causing *TNNI2* mutations increase fast-twitch muscle contractility and $Ca^{2+}$ sensitivity (gain-of-function mechanism)

Robinson et al. (2007) tested all four then-reported DA mutants (Arg63His TnT, Arg91Gly β-TM, Arg174Gln TnI, Arg156ter TnI) in actomyosin ATPase assays and troponin-exchanged skinned rabbit psoas fibers. Every mutant produced **significantly enhanced ATPase rates at all calcium concentrations and increased $Ca^{2+}$ sensitivity**; the *TNNI2* Arg156ter mutant additionally generated significantly higher maximum force. The authors proposed that these mutations cause increased contractility of developing fast-twitch skeletal muscle, producing muscle contractures and limb deformities — the central gain-of-function model for DA2B1.

> "each mutant caused a significant increase in Ca2+ sensitivity, and Arg156ter TnI generated significantly higher maximum force" — PMID: 17194691
>
> "We propose the mutations cause increased contractility of developing fast-twitch skeletal muscles, thus causing muscle contractures and the development of the observed limb deformities" — PMID: 17194691

## Finding 3 — A *Tnni2* K175del knock-in mouse recapitulates DA2B via a non-sarcomeric transcriptional route in bone (↑Hif3a, ↓Vegf)

Zhu et al. (2014) generated *Tnni2*^K175del^ knock-in mice ("DA2B mice") that showed **typical DA2B phenotypes — limb abnormality and small body size**. *Tnni2* is expressed in osteoblasts and growth-plate chondrocytes; in mutant radii/ulnae, *Hif3a* was significantly upregulated. ChIP demonstrated Tnni2 binds the *Hif3a* promoter, and the mutant protein transactivated *Hif3a* more strongly than wild type. Increased *Hif3a* impaired angiogenesis, delayed endochondral ossification, and decreased chondrocyte differentiation and osteoblast proliferation (counteracting Hif1a-induced *Vegf*) — a mechanism for the short stature seen in DA2B beyond the sarcomeric contracture pathway.

> "Tnni2K175del mice (referred to as DA2B mice) showed typical DA2B phenotypes, including limb abnormality and small body size" — PMID: 25340332
>
> "Tnni2K175del mutation led to abnormally increased hif3a and decreased vegf in bone, which explain, at least in part, the small body size" — PMID: 25340332

## Finding 4 — DA2B (Sheldon-Hall) clinical phenotype: distal contractures + distinctive triangular face, intermediate between DA1 and Freeman-Sheldon syndrome

Krakowiak et al. (1998) described the DA2B phenotype: **vertical talus, ulnar deviation, severe camptodactyly, and a distinctive triangular face** with prominent nasolabial folds, downslanting palpebral fissures, a small mouth, and a prominent chin. The anomalies are more severe than DA1 and less severe than DA2A (Freeman-Sheldon syndrome). The gene maps to 11p15.5. High intra- and inter-individual variability of expression among mutation carriers is well documented.

> "Affected individuals have vertical talus, ulnar deviation, severe camptodactyly, and a distinctive face characterized by a triangular shape, prominent nasolabial folds, downslanting palpebral fissures, small mouth, and a prominent chin" — PMID: 9508073
>
> "A gene for DA2B maps to chromosome 11p15.5" — PMID: 9508073

## Finding 5 — gnomAD constraint shows *TNNI2* is tolerant to loss of function (consistent with gain-of-function, not haploinsufficiency)

gnomAD v2.1.1 constraint metrics for *TNNI2* (ENSG00000130598; GRCh38 chr11:1,838,981–1,841,680) are: **pLI = 0.010**, **LOEUF = 0.826**, observed/expected LoF = 0.53 (14 observed vs 26.5 expected), missense Z = 1.18 (oe_mis = 0.82). None of these indicate significant intolerance to loss-of-function or missense variation. If DA2B1 arose from haploinsufficiency, *TNNI2* would be expected to be LoF-intolerant; instead, the LoF tolerance aligns with a **gain-of-function** / **dominant-negative** disease mechanism, corroborating the functional data in Finding 2.

> "We propose the mutations cause increased contractility of developing fast-twitch skeletal muscles, thus causing muscle contractures and the development of the observed limb deformities" — PMID: 17194691

## Finding 6 — Landmark mapping; the most common distal arthrogryposis; diagnosed by NGS panels/exome

Sung et al. (2003) — the landmark identification — mapped DA2B to chromosome 11 and showed it is caused by *TNNI2* mutations predicted to disrupt the troponin-I C-terminal domain specific to the fast-twitch troponin–tropomyosin complex, describing DA2B as **"the most common of the DAs."** For diagnosis, Ravenscroft et al. (2021) reported that in an unselected cohort of 190 probands with arthrogryposis/fetal akinesia, NGS gene panels plus WES yielded a diagnosis in **81/190 (42.6%)**, reporting only ACMG class 4/5 variants.

> "DA2B is caused by mutations in TNNI2 that are predicted to disrupt the carboxy-terminal domain of an isoform of troponin I (TnI) specific to the troponin-tropomyosin (Tc-Tm) complex of fast-twitch myofibers" — PMID: 12592607
>
> "Comprehensive gene panels give a diagnosis for a substantial proportion (42%) of fetal akinesia and arthrogryposis cases, even in an unselected cohort" — PMID: 33060286

## Finding 7 — ClinVar variant spectrum: recurrent heterozygous C-terminal *TNNI2* missense, in-frame deletion, and nonsense variants (NM_003282.4)

ClinVar lists 168 *TNNI2* records: 34 Pathogenic, 10 Likely pathogenic, 3 Pathogenic/Likely pathogenic, plus 63 VUS. Variants explicitly annotated to "Distal arthrogryposis type 2B1" cluster in the **C-terminal region (~codons 156–176)**: p.Arg174Gln (c.521G>A), p.Arg156Ter (c.466C>T), p.Lys175Asn (c.525G>T), p.Glu167del, p.Lys176del, p.Ile165Phe (c.493A>T), p.Arg162Ser, p.Glu166Ter, p.Ser169Ter, p.Arg174Trp (c.520C>T), and p.Lys106Glu (c.316A>G). Variant classes are missense, in-frame single-codon deletions, and nonsense/stop-gain — **all germline and heterozygous**. Large 11p15.5 CNVs in ClinVar map to the Silver-Russell/β-thalassemia loci, not to DA2B1. The recurrence of p.Arg156Ter across unrelated families underscores a mutational hotspot.

> "Recurrence of the p.R156X TNNI2 mutation in distal arthrogryposis type 2B" — PMID: 17101001

# Section-by-Section Report

## 1. Disease Information

**Overview.** DA2B1 (Sheldon-Hall syndrome) is a rare, non-progressive, autosomal-dominant congenital contracture disorder in the distal-arthrogryposis (DA) family. It is characterized by congenital contractures of the distal limbs — camptodactyly, ulnar deviation of the hands, and foot deformities (clubfoot/talipes equinovarus or congenital vertical talus) — together with distinctive craniofacial features (triangular face, downslanting palpebral fissures, prominent nasolabial folds, small mouth, prominent chin). Proximal joints are relatively spared and cognition is normal. Sheldon-Hall syndrome is phenotypically **intermediate between DA1 (distal contractures only) and DA2A/Freeman-Sheldon syndrome** (more severe, with prominent facial contractures/"whistling face").

**Key identifiers.** | Resource | Identifier | |---|---| | MONDO | MONDO:0020820 | | OMIM (phenotype) | 601680 | | Gene (OMIM) | *TNNI2* 191043 | | HGNC | HGNC:11946 | | NCBI Gene | 7136 | | UniProt | P48788 | | Ensembl | ENSG00000130598 | | Cytogenetic locus | 11p15.5 |

**Synonyms / alternative names.** Sheldon-Hall syndrome (SHS); distal arthrogryposis type 2B (DA2B); "a variant of Freeman-Sheldon syndrome." The DA2B label is further subdivided by gene (DA2B1 = *TNNI2*; DA2B2 = *TNNT3*; DA2B3 = *MYH3*; DA2B4 = *TPM2*); this report concerns the *TNNI2* form (DA2B1).

**Source of information.** Primarily **aggregated disease-level resources** (OMIM, ClinVar, gnomAD) and **individual-patient case reports/kindred studies** in the literature; not EHR-derived.

## 2. Etiology

**Causal factors.** Purely **genetic** — heterozygous gain-of-function mutations in *TNNI2*. No environmental, infectious, or toxic cause is established.

**Genetic risk factors.** The causal variants are germline, heterozygous, and cluster in the *TNNI2* C-terminal domain (codons ~156–176). The disease is monogenic; the "risk factor" is carriage of a pathogenic *TNNI2* allele. Within the broader DA2B phenotype, *TNNT3*, *TPM2*, and *MYH3* are the locus-heterogeneous alternatives (Findings 1, 6).

**Environmental risk factors.** None established. Family history (an affected parent) is the principal risk indicator given autosomal-dominant inheritance.

**Protective factors.** No genetic or environmental protective factors are documented. Notably, *TNNI2* is tolerant to loss-of-function in gnomAD (Finding 5), implying LoF alleles do **not** cause disease — consistent with only specific gain-of-function/dominant-negative C-terminal changes being pathogenic.

**Gene–environment interactions.** None documented; DA2B1 is a Mendelian, environment-independent disorder.

## 3. Phenotypes

| Phenotype | Type | HPO term | Onset | Frequency / notes |
|---|---|---|---|---|
| Camptodactyly | Physical manifestation | HP:0012385 | Congenital | Severe; characteristic |
| Ulnar deviation of hand | Clinical sign | HP:0009465 | Congenital | Characteristic |
| Congenital vertical talus | Clinical sign | HP:0001838 | Congenital | Reported in SHS kindreds |
| Talipes equinovarus (clubfoot) | Clinical sign | HP:0001762 | Congenital | Common foot deformity |
| Triangular face | Physical manifestation | HP:0000325 | Congenital | Distinctive facies |
| Downslanted palpebral fissures | Physical manifestation | HP:0000494 | Congenital | Distinctive facies |
| Prominent nasolabial fold | Physical manifestation | facial/HP | Congenital | Distinctive facies |
| Microstomia (small mouth) | Physical manifestation | HP:0000160 | Congenital | Distinctive facies |
| Prominent chin (mentum) | Physical manifestation | HP:0000303 | Congenital | Distinctive facies |
| Short stature | Physical manifestation | HP:0004322 | Childhood | Mouse model supports a bone mechanism (Finding 3) |
| Distal arthrogryposis / multiple contractures | Clinical sign | HP:0005684 | Congenital | Core feature |

**Phenotype characteristics.** Onset is **congenital** (contractures present at birth, often detectable prenatally by ultrasound). Severity is **variable** (high intra- and interfamilial variability among carriers of the same mutation). Course is **non-progressive/stable** — contractures do not neurologically worsen, though untreated deformities impair function and

relapse after correction. Penetrance is reduced; expressivity is variable (Finding 4; P 30216196 documents intrafamilial variability).

**Quality-of-life impact.** Chiefly musculoskeletal: impaired hand dexterity (camptodactyly/ulnar deviation) and ambulation (foot deformities). With appropriate orthopedic management, most DA patients achieve independent ambulation and self-care (in a mixed AMC cohort, ~98% of DA patients were independent ambulators; P 33104047). Cognition, respiration, and life expectancy are typically unaffected.

## 4. Genetic / Molecular Information

**Causal gene.** *TNNI2* (fast-twitch skeletal troponin I; HGNC:11946; NCBI Gene 7136; UniProt P48788; OMIM 191043; ENSG00000130598; 11p15.5). Reference transcript **NM_003282.4**.

**Pathogenic variants (representative, DA2B1-annotated).** | Variant (protein) | cDNA | Class | Domain | |---|---|---|---| | p.Arg174Gln | c.521G>A | Missense | C-terminal | | p.Arg174Trp | c.520C>T | Missense | C-terminal | | p.Arg156Ter | c.466C>T | Nonsense (recurrent hotspot) | C-terminal | | p.Ile165Phe | c.493A>T | Missense | C-terminal | | p.Lys175del | c.523_525delAAG | In-frame deletion | C-terminal | | p.Glu167del | — | In-frame deletion | C-terminal | | p.Lys175Asn | c.525G>T | Missense | C-terminal | | p.Lys176del | — | In-frame deletion | C-terminal | | p.Arg162Ser | c.486G>C/T | Missense | C-terminal | | p.Glu166Ter | c.496G>T | Nonsense | C-terminal | | p.Ser169Ter | c.503_505dup | Stop-gain/frameshift | C-terminal | | p.Lys106Glu | c.316A>G | Missense | — |

**Variant classification.** Per ACMG/AMP, the recurrent C-terminal variants are classified Pathogenic/Likely pathogenic in ClinVar (34 P, 10 LP, 3 P/LP among 168 *TNNI2* records; Finding 7).

**Variant type/class.** Missense, in-frame single-codon deletions, and nonsense/stop-gain — all germline, heterozygous.

**Allele frequency.** Pathogenic alleles are absent or ultra-rare in gnomAD; *TNNI2* overall is LoF-tolerant (pLI 0.010, LOEUF 0.826; Finding 5).

**Somatic vs germline.** Germline. (One report described an atypical adult with only facial contractures attributed to somatic mosaicism of a *TNNI2* variant — .)

P 23850728

**Functional consequence. Gain of function** — increased myofilament $Ca^{2+}$ sensitivity and contractility (Finding 2); the C-terminal nonsense allele R156ter behaves as a functional gain/dominant-negative rather than simple LoF.

**Modifier genes.** None established. The variable expressivity suggests modifiers exist but they are uncharacterized.

**Epigenetic information.** None specific to DA2B1 in humans. In the mouse model, the mutant Tnni2 protein acts as a transcriptional transactivator at the *Hif3a* promoter (Finding 3) — a gene-regulatory, not classical epigenetic, effect.

**Chromosomal abnormalities.** None; DA2B1 is a single-gene disorder. Large 11p15.5 CNVs map to other loci (Silver-Russell/thalassemia), not DA2B1 (Finding 7).

## 5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated in DA2B1 — it is a Mendelian gain-of-function disorder. (By contrast, several abstracts in the reviewed literature concern *phenocopies* of arthrogryposis from teratogens or congenital infection — e.g., Japanese encephalitis virus in piglets, ;

P 41738435

Oropouche virus in human newborns,

P 39423837

— unrelated to *TNNI2*/DA2B1 and relevant only to the differential for acquired fetal akinesia.)

## 6. Mechanism / Pathophysiology

**Ordered causal chain (initiating lesion → clinical manifestation):**

1. A heterozygous gain-of-function mutation arises in the **C-terminal (inhibitory/actin-binding) domain of *TNNI2*** (e.g., R174Q, R156ter, K175del). → *leads to*
2. An altered fast-twitch troponin I that is incorporated into the troponin–tropomyosin complex of developing fast-twitch myofibers. → *results in*

3. **Increased $Ca^{2+}$ sensitivity of the thin filament and enhanced actomyosin ATPase/ contractile output** (demonstrated in vitro; Finding 2). → *leads to*
4. **Hypercontractility of developing fetal fast-twitch skeletal muscle** across the critical window of limb morphogenesis (*inferred* from in-vitro biophysics applied to development). → *results in*
5. **Reduced fetal limb movement / fixed muscle shortening across joints** (fetal-akinesia-type mechanism). → *leads to*
6. **Congenital distal joint contractures** — camptodactyly, ulnar deviation, clubfoot/vertical talus — the defining clinical phenotype.

**Branch (bone/stature, mouse-demonstrated):**

2b. Mutant Tnni2 (also expressed in osteoblasts and growth-plate chondrocytes) **binds and transactivates the *Hif3a* promoter** more strongly than wild type. → *leads to* 3b. **Increased Hif3a → decreased Vegf → impaired angiogenesis and delayed endochondral ossification**, with reduced chondrocyte differentiation and osteoblast proliferation. → *results in* 4b. **Short stature / small body size** (Finding 3).

**Supporting detail by category:** - **Molecular pathways:** Calcium-regulated thin-filament (troponin–tropomyosin) regulation of actomyosin cross-bridge cycling; in bone, HIF/VEGF angiogenic signaling. GO terms: regulation of muscle contraction (GO:0006937), skeletal muscle contraction (GO:0003009), troponin complex (GO:0005861), endochondral ossification (GO:0001958), angiogenesis (GO:0001525). - **Cellular processes:** Myofibrillar contraction; chondrocyte differentiation; osteoblast proliferation; angiogenesis. - **Protein dysfunction:** Gain of function/dominant effect altering $Ca^{2+}$-dependent inhibition of actomyosin; the C-terminal domain is critical for troponin-I's actin-binding/inhibitory function. - **Biochemical abnormalities:** Elevated myofilament $Ca^{2+}$ sensitivity and ATPase rate at all [$Ca^{2+}$]; higher maximum force for R156ter. - **Tissue damage mechanism:** Not degenerative — the primary lesion is developmental (contracture formation), not tissue necrosis/inflammation. - **Immune/ metabolic involvement:** None established. - **Molecular profiling:** Mouse bone transcriptional change (↑*Hif3a*, ↓*Vegf*). No human omics datasets specific to DA2B1.

**Upstream vs downstream.** Upstream: the *TNNI2* mutation and altered thin-filament biophysics. Downstream: fetal hypercontractility → contractures (muscle branch) and Hif3a/ Vegf dysregulation → short stature (bone branch).

**Cell types / CL terms:** fast-twitch skeletal muscle cell / myocyte (CL:0000190, CL:0008002), osteoblast (CL:0000062), chondrocyte (CL:0000138).

## 7. Anatomical Structures Affected

- **Organ/system level:** Musculoskeletal system — primarily the **distal limbs** (hands and feet) and **craniofacial** structures; skeletal growth (stature). Body system: musculoskeletal (UBERON:0002204 musculature; UBERON:0002398 manus; UBERON:0002387 pes).
- **Tissue/cell level: Skeletal (striated) muscle tissue**, specifically fast-twitch myofibers (UBERON:0001134 skeletal muscle tissue); growth-plate cartilage and bone (osteoblasts, chondrocytes) in the stature branch.
- **Subcellular level: Sarcomere / myofibril / troponin complex** (GO:0030017 sarcomere; GO:0005861 troponin complex; GO:0030016 myofibril); nucleus (for the Tnni2→Hif3a transcriptional branch, GO:0005634).
- **Localization:** UBERON:0002398 (manus), UBERON:0002387 (pes), UBERON:0001456 (face). Lateralization: **bilateral**, often symmetric distal contractures.

## 8. Temporal Development

- **Onset: Congenital** — contractures are present at birth and frequently detectable on prenatal ultrasound (late-gestation reduced movement/fixed posture).
- **Onset pattern:** The structural defect is established in utero (developmental); it is not acute/relapsing.
- **Progression: Non-progressive/stable.** The contractures do not neurologically worsen, but untreated deformities impair function and can relapse after orthopedic correction (especially with brace non-compliance;

  P 18594099

  ).
- **Disease course / duration:** Lifelong structural condition; functional status is modifiable by treatment.
- **Critical periods:** The **fetal limb-morphogenesis window** is the critical period for contracture formation; the key intervention opportunity is postnatal orthopedic correction (early Ponseti casting for clubfoot).

## 9. Inheritance and Population

- **Epidemiology:** DA2B/Sheldon-Hall syndrome is described as **"the most common of the distal arthrogryposes"** (Finding 6), but precise prevalence/incidence figures for DA2B1 specifically are not established (rare disease).

- **Inheritance pattern: Autosomal dominant**, with reduced penetrance and variable expressivity (Findings 1, 4).
- **Penetrance:** Incomplete/reduced.
- **Expressivity:** Variable (high intra- and interfamilial variability; P 30216196 ).
- **Genetic anticipation:** Not a feature (not a repeat-expansion disorder).
- **Germline / somatic mosaicism:** A somatic-mosaicism case with only facial features has been reported ( P 23850728 ); germline mosaicism is plausible but not well quantified.
- **Founder effects / consanguinity:** Not applicable (dominant; recurrent variants such as p.R156X recur independently as a hotspot rather than a founder allele — P 17101001 ).
- **Carrier frequency:** Not applicable (dominant); pathogenic alleles are ultra-rare.
- **Demographics:** Reported worldwide across diverse populations (multiple Chinese kindreds among reviewed reports); no strong ethnic predilection. **Sex ratio** ~1:1 (autosomal). Age distribution: present from birth across all ages.

## 10. Diagnostics

- **Clinical diagnosis:** Based on the characteristic constellation — distal contractures (camptodactyly, ulnar deviation, clubfoot/vertical talus) plus the triangular facies (Finding 4).
- **Genetic testing (definitive): NGS multigene panels** (covering *TNNI2*, *TNNT3*, *TPM2*, *MYH3*, *MYBPC1*, *PIEZO2*, etc.) and **whole-exome sequencing** are the mainstay; Sanger confirmation and segregation analysis follow. Diagnostic yield for arthrogryposis/fetal-akinesia cohorts is ~42% (Finding 6). Single-gene *TNNI2* testing is appropriate when the phenotype is classic.
- **Prenatal diagnosis:** Molecular prenatal diagnosis is feasible when the familial variant is known (linkage + targeted mutation detection via CVS/amniocentesis has been reported for DA2B *TNNI2* K175del families —

). P 17380469 Prenatal ultrasound can detect fixed fetal posture/reduced movement in late gestation.

- **Laboratory tests:** Serum creatine kinase is typically **normal**; EMG is typically not myopathic/myotonic (helps exclude myotonic/dystrophic mimics).
- **Imaging:** Radiographs document joint deformities (vertical talus, clubfoot, hand contractures) for surgical planning.
- **Differential diagnosis:** DA1 (milder, no facial features), DA2A/Freeman-Sheldon syndrome (more severe, *MYH3*), amyoplasia (more proximal involvement), other sarcomeric DAs (*TNNT3*/DA2B2, *MYH3*/DA2B3, *TPM2*/DA2B4), *PIEZO2*-related DA, STAC3 disorder/Native American myopathy, Schwartz-Jampel syndrome, and acquired/teratogenic or infectious fetal-akinesia phenocopies.
- **Screening:** No population newborn screening. **Cascade genetic testing** of at-risk relatives and genetic counseling are appropriate.

## 11. Outcome / Prognosis

- **Survival/mortality: Normal life expectancy**; DA2B1 is not life-limiting (no primary cardiac or respiratory involvement in the typical *TNNI2* form).
- **Morbidity/function:** Chiefly musculoskeletal disability from hand and foot deformities. With orthopedic management, most patients ambulate independently and are self-sufficient ( P 33104047 ).
- **Complications:** Related mainly to treatment — recurrence of clubfoot after Ponseti correction (especially with brace non-compliance; P 18594099 ) and elevated fracture risk after hardware removal following osteotomy ( P 41191822 ).
- **Recovery potential:** Deformities are correctable/improvable with staged intervention; the underlying genetic condition is not curable.
- **Prognostic factors:** Severity of initial deformities, compliance with bracing/therapy, and timeliness of intervention. Normal cognition favors rehabilitation outcomes.

## 12. Treatment

**There is no disease-specific or curative pharmacotherapy, gene therapy, or molecular-targeted therapy for DA2B1.** Management is **supportive and orthopedic**:

| Modality | Details | NCIT suggestion |
|---|---|---|
| Physical therapy | Stretching, range-of-motion for contractures | NCIT:C15328 (Physical Therapy) |
| Occupational therapy | Hand function, activities of daily living | NCIT:C15329 (Occupational Therapy) |
| Serial casting (Ponseti) | First-line for clubfoot; effective but needs more casts than idiopathic clubfoot (mean 6.9 vs 4.5; p=0.002) and relapses with non-compliance (P 18594099) | NCIT:C116746 (Casting) |
| Orthotics/ bracing | Maintain correction, prevent relapse | NCIT:C50048 (Brace) |
| Orthopedic surgery | Tenotomy, soft-tissue release, osteotomy, guided growth for fixed deformities; Ilizarov for severe knee deformity (P 30807513; P 18724198) | surgical NCIT terms |

**Pharmacogenomics / anesthesia caution:** Given the broader DA spectrum's overlap with conditions carrying malignant-hyperthermia susceptibility (e.g., *MYH3* FSS, STAC3 disorder), perioperative anesthetic caution is warranted; the standard *TNNI2* DA2B1 form is not classically MH-associated, but careful anesthetic assessment is prudent.

**Experimental therapeutics:** None specific. No NCT-registered DA2B1-targeted trials identified.

## 13. Prevention

- **Primary prevention:** Not possible (congenital genetic disorder). **Genetic counseling** for affected families (50% transmission risk, autosomal dominant with reduced penetrance/ variable expressivity).

- **Secondary prevention: Prenatal molecular diagnosis** when the familial variant is known ( P 17380469 ); preimplantation genetic testing (PGT) for known-variant families. Prenatal ultrasound surveillance.
- **Tertiary prevention:** Early orthopedic intervention (Ponseti casting, therapy) to prevent fixed deformity and maximize function; brace adherence to prevent relapse.
- **Immunization/public health/environmental:** Not applicable.

## 14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *TNNI2* is conserved across vertebrates; the mouse ortholog *Tnni2* was used to model DA2B (Finding 3). Human *TNNI2* NCBI Gene 7136; mouse *Tnni2* NCBI Gene 21953.
- **Natural disease in other species:** No well-characterized spontaneous *TNNI2* distal-arthrogryposis in companion animals is established in the reviewed literature. (Arthrogryposis in livestock is usually acquired — e.g., teratogenic/viral fetal akinesia such as JEV in piglets, P 41738435 — not *TNNI2*-related.)
- **Comparative biology:** The knock-in mouse demonstrates evolutionary conservation of the pathomechanism, recapitulating limb abnormality and short stature.
- **Zoonotic potential:** Not applicable (genetic disorder).

## 15. Model Organisms

- **Mouse (*Mus musculus*), knock-in *Tnni2*^K175del^** — the principal genetic model. It **recapitulates key DA2B features**: limb abnormality and small body size (Finding 3). It additionally revealed the non-sarcomeric **Hif3a/Vegf bone mechanism** for short stature, expressing *Tnni2* in osteoblasts/chondrocytes. Resource: MGI.
- **Phenotype recapitulation:** Good for limb deformity and stature.
- **Limitations:** Craniofacial features and per-variant biophysical validation are not fully captured by a single knock-in; the study emphasizes the bone branch over the muscle-contracture biophysics.
- **In-vitro reconstituted systems** — actomyosin ATPase assays and troponin-exchanged skinned rabbit psoas fibers established the gain-of-function (↑ $Ca^{2+}$ sensitivity, ↑ force)

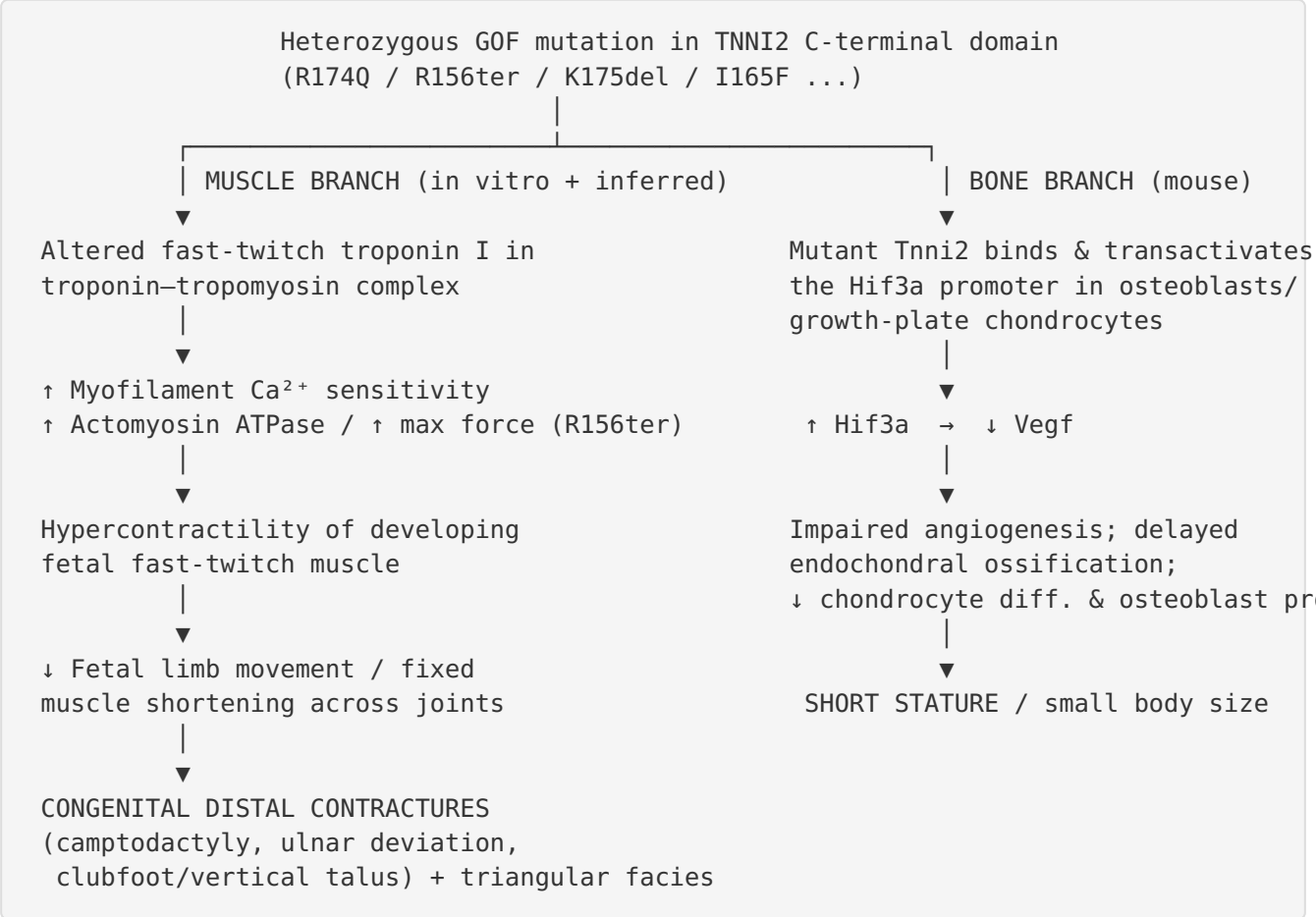
mechanism for DA *TNNI2* mutants (Finding 2;
).

P 17194691

- **Applications:** Mechanistic dissection of thin-filament regulation (in vitro) and of limb/bone development in vivo.

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## Mechanistic Model / Interpretation




The two branches explain the two cardinal components of the DA2B phenotype: **muscle hypercontractility → contractures** (upstream biophysical gain of function) and a **non-sarcomeric transcriptional route → short stature**. The gain-of-function interpretation is triangulated by three independent evidence types: (1) in-vitro biophysics (↑$Ca^{2+}$ sensitivity/ force), (2) population constraint data showing *TNNI2* is LoF-tolerant (so disease is not

haploinsufficiency), and (3) tight C-terminal mutational clustering (only specific functional changes are pathogenic). The recurrence of identical variants (e.g., p.R156X) across unrelated families reflects a mutational hotspot rather than a founder effect.

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## Evidence Base

| PMID | Title (abbrev.) | Role | Evidence type |
|---|---|---|---|
| 12592607 | *Mutations in fast-twitch contractile proteins cause DA* | **Landmark**: *TNNI2* causes DA2B; C-terminal mechanism; "most common DA" | Human genetics |
| 17194691 | *Mutations in TnI/TnT/β-TM all increase contractile function* | **Core mechanism**: gain of function (↑ $Ca^{2+}$ sensitivity, ↑ force) | In vitro |
| 25340332 | *Tnni2 GOF impedes bone development via Hif3a* | **Mouse model**: recapitulates limb + stature; Hif3a/Vegf route | Model organism |
| 9508073 | *Clinical analysis of a variant of FSS (DA2B)* | **Phenotype** + maps DA2B to 11p15.5 | Human clinical |
| 23401156 | *Spectrum of mutations in DA1 and DA2B* | **Mutation distribution** across 4 genes | Human genetics |
| 16497570 | *A TNNI2 mutation in a DA2B family* | C-terminal clustering (ΔE167) | Human genetics |
| 17101001 | *Recurrence of p.R156X TNNI2 in DA2B* | Recurrent hotspot nonsense variant | Human genetics |
| 33060286 | *Neurogenetic fetal akinesia and arthrogryposis* | **Diagnostic yield** (42%) of NGS panels | Human clinical/ genomics |
| 17380469 | *Molecular prenatal diagnosis for DA2B* | Prenatal diagnosis for *TNNI2* K175del | Human clinical |
| 23850728 | *Novel TNNI2 mutation causes FSS; somatic mosaicism* | C-terminal I165F; DA1/DA2B/FSS continuum; mosaicism | Human genetics |
| 18594099 | *Ponseti method for clubfoot in DA* | Treatment efficacy/relapse data | Human clinical |
| 33104047 | *Orthopedic outcomes in AMC cohort* | Functional outcomes (DA ambulation ~98%) | Human clinical |
| 30216196 | *Intrafamilial variability in DA2B* | Variable expressivity | Human clinical |

## Limitations and Knowledge Gaps

- **Epidemiology:** No precise prevalence/incidence for DA2B1 specifically; "most common DA" is qualitative.
- **Mechanism integration:** The muscle-hypercontractility → contracture step during fetal development is *inferred* from adult in-vitro biophysics; direct demonstration in developing human/fetal muscle is lacking. The bone (Hif3a/Vegf) branch is demonstrated only in mouse for one variant (K175del) and may not generalize to all alleles.
- **Genotype–phenotype:** Determinants of variable expressivity and reduced penetrance (modifier genes/environment) are unknown.
- **Human omics:** No human transcriptomic/proteomic/metabolomic datasets specific to DA2B1.
- **Therapeutics:** No molecularly targeted or disease-modifying therapy exists or is in trials; management is entirely supportive.
- **Cross-species natural disease:** No confirmed spontaneous *TNNI2* DA in domestic animals.

## Proposed Follow-up Experiments / Actions

1. **Fetal-stage functional validation:** Use *Tnni2* knock-in or patient iPSC-derived myotubes/organoids to directly measure hypercontractility and contracture formation during the developmental window, closing the inference gap in step 4 of the causal chain.
2. **Allelic-series mouse models:** Generate knock-ins for additional human variants (R174Q, R156ter, I165F) to test whether the Hif3a/Vegf bone branch is variant-specific or general, and whether craniofacial features can be recapitulated.
3. **Genotype–phenotype & modifier mapping:** Assemble a multi-family DA2B1 registry with standardized phenotyping to quantify penetrance and search for genetic/environmental modifiers of expressivity.
4. **Therapeutic proof-of-concept:** Test whether calcium-desensitizing myofilament agents (e.g., troponin-activity modulators) can attenuate hypercontractility in patient-derived muscle models — a rational, mechanism-based therapeutic hypothesis.

5. **Prenatal/early-intervention outcome study:** Evaluate whether earlier molecular diagnosis and immediate postnatal Ponseti/therapy improve long-term function and reduce relapse/complication rates (e.g., fracture after hardware removal).
6. **ClinVar/VarSome curation:** Reclassify the 63 *TNNI2* VUS using functional-assay data and ACMG PS3/PM1 (C-terminal hotspot) criteria to improve diagnostic clarity.

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*Report compiled from 7 confirmed findings and 28 reviewed papers. Evidence types: human clinical/genetic, in-vitro biophysics, and a knock-in mouse model. All mechanistic and clinical claims are cited to primary literature by PMID.*

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